

TrustEpisode: Deterministic Episode Formation and Manifest-Scoped Calibration for Partial EDR/NDR Telemetry

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ABSTRACT

Multi-step attacks appear as temporally dispersed endpoint and network observations rather than one decisive alert. Correlation must therefore define which events form a candidate episode, preserve deterministic revisions under late data, and distinguish an unsupported probability from a low probability when telemetry is missing.

We present TRUSTEPISODE, a label-blind EDR/NDR pipeline with explicit ownership boundaries. Evidence Fabric emits canonical events and independent source-health intervals; Episode Synthesis alone owns membership and immutable revisions; a two-feature stateless scorer emits a raw logit; and a manifest-scoped affine calibrator emits a nullable maliciousness probability. A separate deterministic evaluator reports a five-axis evidence-sufficiency profile without changing that probability. An AuditRecord Assembler creates one terminal, append-only outcome per revision: a complete AuditRecord from independently produced inputs, or an explicit AssemblyFailureRecord without fabricated score or probability fields.

We specify an end-to-end validation protocol in which CALDERA executes seven multi-step attack families on isolated live Ubuntu systems observed through EDR/NDR collection paths. Paired event-loss, delay, reordering, clock-shift, duplication, source-masking, and physical-outage conditions test both performance degradation and transitions in the five evidence-sufficiency axes. The analysis table and robustness plots remain pending until the registered executions complete; no synthetic performance values or confidence intervals are reported.

KEYWORDS

attack episode formation, probability calibration, partial telemetry, provenance, EDR, NDR

1 INTRODUCTION

Multi-step attacks rarely produce one conclusive alert. A remote connection, script execution, file transfer, or administrative action can be benign alone yet material when connected by time, entity, session, and process relations. Endpoint detection and response (EDR) and network detection and response (NDR) also fail differently: an event may be absent because no action was observed, because no detector fired, or because a collector or parser was unavailable. Treating all three states as zero evidence silently changes both correlation and probability semantics.

Existing alert correlation and provenance systems reconstruct causal context, but their candidate boundaries, late-data behavior, and calibration contracts are often not explicit enough for byte-identical replay [2, 7–9]. Conversely, an anomaly or severity score is not automatically a probability, and calibration is meaningful only for the candidate population and pipeline version on which it was validated [4, 5]. These gaps are coupled: changing episode

membership changes the scored population, while hiding source failure behind a zero feature can make an unsupported mapping appear confidently benign.

TRUSTEPISODE addresses this problem with a deliberately narrow contract. It transforms only EDR/NDR observations into deterministic EpisodeRevisions, computes a compact raw logit from detector deviation and independent corroboration, emits a calibrated probability only under an exact manifest match, reports evidence sufficiency independently, and assembles one canonical terminal assembly outcome per revision. Ground truth is unavailable to every online producer and is joined only after inference.

Contributions. This paper makes four contributions:

- (1) a single-owner online object model and deterministic episode state machine with fixed guards, total ordering, watermark, and revision policy;
- (2) a compact scorer plus manifest-scoped affine calibration that returns canonical null outside validated support;
- (3) explicit detector-availability and five-axis sufficiency truth tables, together with a schema-ready AuditRecord contract with closed types, nullability, and digest sets; and
- (4) a CALDERA-based validation protocol on isolated live Linux systems, with fixed scenario cards, paired telemetry perturbations, physical outages, sufficiency-transition analysis, matching, and run-level statistics.

2 RELATED WORK

Alert correlation and provenance. Classical alert-correlation methods connect observations into attack scenarios using temporal, causal, and prerequisite relations [9]. Provenance-based intrusion detection preserves process and information-flow context; HOLMES, NODLINK, and KAIROS show how causal graphs support multi-stage reconstruction and investigation [2, 7, 8]. CAPTAIN further demonstrates that a lightweight, interpretable rule system can adapt deployment-specific parameters [14]. TRUSTEPISODE differs by treating deterministic cross-source candidate construction, revision ownership, and candidate-conditioned probability calibration as one versioned contract rather than presenting a new graph detector.

Calibration and incomplete telemetry. Ranking and calibration answer different questions: a high AUPRC does not imply reliable probabilities [4, 5]. Platt-style affine calibration is compact and ranking-preserving when its slope is positive, but only for a fixed candidate population and support regime [11]. Missing-modality research likewise shows that absent inputs require explicit handling rather than silent zero imputation [16]. TRUSTEPISODE therefore separates observed source masks, authenticated health, detector availability, calibration status, and sufficiency. The raw scorer cannot read health or labels, and unsupported records retain a diagnostic logit but no numeric probability.

Scope. The method does not model operational impact, response priority, autonomous response, CTI enrichment, or language-agent behavior. Its claims are limited to deterministic EDR/NDR episode formation, manifest-scoped maliciousness calibration, explicit evidence sufficiency, and replayable audit records.

3 TRUSTEPISODE METHOD

TRUSTEPISODE converts EDR/NDR telemetry into deterministic candidate revisions and version-scoped evidence records. Evidence Fabric and Health Monitor respectively own events and source health; Episode Synthesis alone owns membership; Feature Extractor emits health-blind D, C, m and detector observations; Stateless Scorer emits z ; Scope Resolver emits resolved detector availability, κ , the calibration-group key, and decision_eligible; Calibrator emits nullable P ; and Sufficiency Evaluator emits structured U . For each revision, the AuditRecord Assembler emits exactly one terminal AssemblyOutcome: a canonical AuditRecord when its join succeeds, or an immutable AssemblyFailureRecord when it cannot safely assemble one. Ground truth is joined only after inference. The method estimates malicious activity inside a frozen EDR/NDR candidate; it does not estimate enterprise risk, impact, response priority, or unobservable attacks. Figure 1 shows these boundaries.

3.1 Problem Scope, Output Contract, and Ownership

Let C contain every emitted EpisodeRevision and define

$$C_{\text{sup}} = \{\epsilon^{(r)} \in C : \kappa_{\epsilon^{(r)}} \in \{\kappa_G, \kappa_g\}\}, \quad (1)$$

where κ_G, κ_g , and $\kappa_\emptyset$ serialize as supported_global, supported_group, and out_of_support. The offline target $Y_{\epsilon^{(r)}} \in \{0, 1\}$ is revision specific: evidence first accepted in $r+1$ cannot relabel r . Positive and negative revisions form the primary supervised cohort; mixed and ambiguous revisions enter only the registered sensitivity analysis.

When defined, the candidate-conditioned output is

$$P_{\epsilon^{(r)}} = \Pr(Y_{\epsilon^{(r)}} = 1 \mid \epsilon^{(r)}, v_{\text{episode}}, v_{\text{feature}}, v_{\text{model}}, v_{\text{cal}}). \quad (2)$$

It is not the unconditional probability that an organization is under attack. Online schemas are label blind and reject scenario, marker, window, and label fields. Source-health messages are authenticated; activity outside observable EDR/NDR traces is outside the claim.

Table 1 assigns every online object and decision field to one producer and forbids downstream components from silently taking over that ownership.

3.2 Evidence, Health, and Detector Availability

Evidence Fabric maps each source record to a NormalizedEvent with stable ID, event and ingest times, source instance, canonical subject/object/action, immutable raw pointer, and root dependency group. Forwarded copies retain their root and cannot add evidence. OCSF v1.8.0 is only a field-alignment target [10]. Health Monitor independently emits authenticated SourceCoverage intervals from heartbeat, lag, parser, and transport observations; silence is never interpreted as failure. Each manifest fixes a 30 s heartbeat

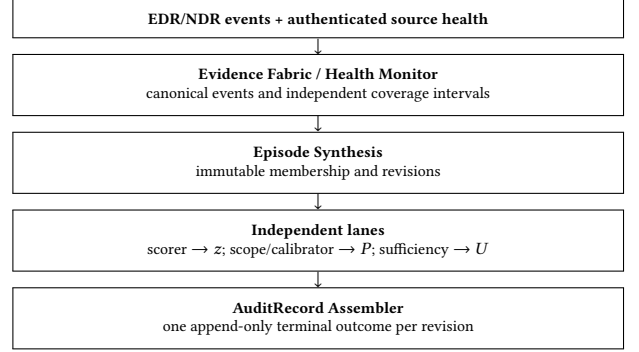


Figure 1: Label-blind online and offline boundaries. The primary candidate path contains EDR/NDR only. Every arrow enters a declared producer; the AuditRecord Assembler owns the terminal AssemblyOutcome and the append-only store is only a sink. U has no numerical path into the scorer or calibrator.

period: healthy requires no missed heartbeat, lag ≤ 120 s, valid parser, and live transport; degraded means one missed heartbeat or $120 < \text{lag} \leq 600$ s while transport remains live; down requires an authenticated collector/transport stop or two consecutive missed heartbeats; missing or conflicting intervals yield unknown.

For $s \in \mathcal{S}^* = \{\text{EDR}, \text{NDR}\}$,

$$m_{\epsilon, s} = 1[\epsilon \text{ contains an observed event from } s]. \quad (3)$$

Feature Extractor is the sole producer of m and health-independent observation $o_{\epsilon, s}^{\text{det}} \in \{\text{detected}, \text{examined_none}, \text{not_examined}\}$. It reads EpisodeRevision and detector outputs, but never SourceCoverage. Scope Resolver is the sole producer of resolved availability a^{det} and selects coverage at $t_{\text{sup}} = \text{EpisodeRevision.event_time_end}$, not the later watermark or assembly wall time. It requires exactly one content-distinct covering interval per expected source instance and applies, in order: missing or conflicting intervals $\mapsto$ unknown; unexpected/down source $\mapsto$ collector_unavailable; invalid parser $\mapsto$ parser_invalid; unknown health or not_examined $\mapsto$ unknown; otherwise detected $\mapsto$ available and examined_none $\mapsto$ no_detection. Thus event silence never becomes health. Finally, $q_{\epsilon, s} = 1$ requires lineage-deduplicated detector support tied to a core entity or edge. Scope Resolver consumes $(m, o_{\epsilon, s}^{\text{det}}, h)$, eligible-detector count n , and exact versions and alone emits SupportDecision. It contains $a^{\text{det}}, \kappa, g, e, n$, an eligibility reason, coverage_ref_ids, and health_snapshot_digest; e is decision_eligible. Before calibration resolution, $n = 0$ deterministically yields $\kappa = \kappa_\emptyset, e = 0$, reason no_eligible_detector_value, and therefore JSON-null P , even when both resolved states are no_detection. Otherwise $e = 1$ iff all support predicates pass and $\kappa \in \{\kappa_G, \kappa_g\}$. Coverage IDs identify the exact EDR/NDR SourceCoverage objects used by the resolver. The digest covers their RFC 8785 canonical health snapshot.

Table 1: Single-owner online object contract.

Object	Sole producer	Consumers	Forbidden behavior
NormalizedEvent SourceCoverage	Evidence Fabric Health Monitor	Episode Synthesis Scope Resolver; Sufficiency Evaluator; Assembler	label, scenario, or health fields infer outage from event silence; independently resample health
EpisodeRevision / LateEvidenceRecord	Episode Synthesis	Feature Extractor; Sufficiency; Assembler / late store	rewrite a revision; admit beyond-horizon evidence
FeatureVector $[D, C, m, o^{det}]$ Raw logit z SupportDecision Nullable P Structured U AssemblyOutcome	Feature Extractor Stateless Scorer Scope Resolver Calibrator Sufficiency Evaluator AuditRecord Assembler	Scorer; Scope/Sufficiency; Assembler Calibrator; Assembler Calibrator; Sufficiency; Assembler Assembler; downstream review Assembler; downstream review Append-only store; replay; evaluation	SourceCoverage, inferred health, or labels probability wording test-selected status; second eligibility/health producer modification by U ; numeric unsupported output scalarization or recomputation of P both success and failure for one revision; update/delete

3.3 Deterministic Episode Synthesis and Revisions

For event e_i and open episode E , eligibility is

$$\ell(e_i, E) = 1[\Delta_t \leq 300s \wedge R = 1 \wedge \text{span}(E \cup \{e_i\}) \leq 3600s \wedge H = 0]. \quad (4)$$

R comprises process-causal, authenticated-session, endpoint-mapped connection, and specific-entity relations. Their precedence ρ is respectively 4, 3, 2, 1; n_R is the number of distinct satisfied relation classes. $H = 1$ when all support comes from an entity with at least 25 distinct relations in the preceding 600 s. Eligible candidates use the sole order $(-\rho, -n_R, \Delta_t, t_{start}, \text{episode_id})$.

Zero/one candidates start/attach. Multiple candidates merge only when there are at most three parents, every parent is connected through the new event by a causal, session, or endpoint relation, and the result has at most 500 events and 200 entities; otherwise the first ordered candidate receives the event. Start and merge IDs hash the policy version and canonical inputs; membership order is $(t^{event}, t^{ingest}, \text{source_instance}, \text{event_id})$.

The watermark is maximum observed ingest time minus 120 s. When an open lineage first satisfies the watermark, Episode Synthesis emits a finalized successor and freezes t_f and deadline $t_f + 900$ s. The lineage leaves the open-candidate index but remains in a separate revisable index. A closed-lineage index retains canonical relation keys and the latest revision until deadline plus $W_{ret} = 86400$ s. Before that fixed deadline, a late event is accepted only if Equation 4 holds against the latest revision; acceptance emits a new finalized successor with the unioned membership and the same t_f and deadline. It does not reopen the lineage or extend the horizon. At the deadline a membership-identical closed successor is emitted, removed from the revisable index, and retained in the closed-lineage index. A later related event emits an immutable LateEvidenceRecord with reason beyond_revision_horizon, closed_lineage_match, or duplicate_after_terminal_outcome to the late store; it cannot create an EpisodeRevision or change membership, z , P , or an earlier AuditRecord. An event older than W_{ret} is recorded as closed_index_expired; it cannot start a fresh episode.

Algorithm 1: label-blind update. Validate and root-deduplicate the event. First query revisable and closed-lineage indexes using event time and canonical relation keys. If any match, select one by $(\text{state priority}, -\rho, -n_R, \Delta_t, t_{start}, \text{id})$ with revisable before closed; emit exactly one accepted successor or LateEvidenceRecord and return. Only when no non-open lineage matches may the event query sorted open candidates and execute exactly one

start/attach/merge action. Events past W_{ret} fail ingress. Advance the watermark and emit each immutable state successor or side record. Replay asserts equality of IDs, membership, links, states, late records, features, outputs, evidence references, and canonical digests. Figure 2 illustrates the fixed transition points.

3.4 Compact Features and Stateless Raw Score

The primary scorer input is

$$x_\epsilon = [D_\epsilon, C_\epsilon]. \quad (5)$$

The frozen detector registry contains exactly three EDR IDs and three NDR IDs; each emits only a finite binary64 value $r \in [0, 1]$, with larger values more suspicious. Source family, detector ID, and dependency group form the deduplication key. Byte-identical retries collapse; a conflicting value or reference rejects the FeatureVector. For each detector, negative training revisions define $\hat{F}(r) = (\#\{v < r\} + 0.5\#\{v = r\})/N$; $N = 0$ invalidates the registry. Sort normalized values descending, breaking ties by UTF-8 detector ID then dependency-group ID. If $n > 0$, D is the mean of the first $\min(3, n)$ values; if $n = 0$, $D = 0$ is a diagnostic placeholder and the revision is not decision eligible. Missing/nonfinite/out-of-range values are never zero-imputed. For source s , $q_{\epsilon, s} = 1$ iff at least one root-deduplicated registered detector has detected, a legal value, and a root tied to a core entity or selected edge; otherwise it is zero. Thus corroboration is fixed to

$$C_\epsilon = q_{\epsilon, \text{EDR}} q_{\epsilon, \text{NDR}}, \quad K = 2. \quad (6)$$

With $\lambda = 10^{-4}$, the stateless scorer is the unique frozen solution

$$\begin{aligned} (\hat{b}, \hat{w}_D, \hat{w}_C) = \arg \min_{b \in \mathbb{R}, w_D, w_C \geq 0} & \frac{1}{N} \sum_i [\log(1 + e^{z_i}) - y_i z_i] \\ & + \frac{\lambda}{2} (b^2 + w_D^2 + w_C^2), \end{aligned} \quad (7)$$

$$z_i = b + w_D D_i + w_C C_i.$$

Its cohort is the first finalized revision per training lineage with exact EDR/NDR health/parser support and an offline positive/negative label; mixed, ambiguous, OOS, and failed outcomes are excluded and counted. Deterministic binary64 L-BFGS-B starts at zero, uses stable logaddexp, stops at projected-gradient infinity norm $\leq 10^{-9}$ and objective change $\leq 10^{-12}$, and fails closed after 10,000 iterations. The RFC 8785/SHA-256 model artifact covers ordered training revision IDs/labels, detector-registry and normalizer digests, matrix bytes, λ , solver contract, and final coefficients. Health, m , a^{det} , κ , U , and outage status never enter x_ϵ .

Deterministic episode and revision lifecycle

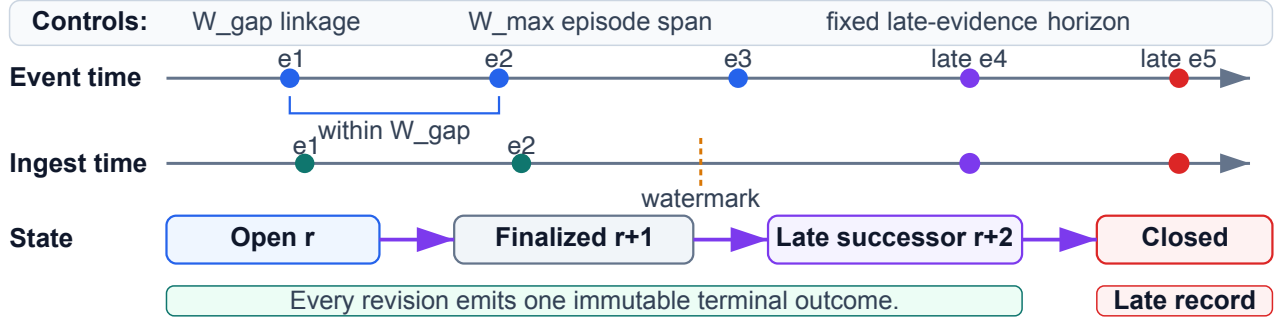


Figure 2: Deterministic lifecycle. Event time controls linking, ingest time controls the 120 s watermark, accepted evidence within 900 s creates a successor revision, and evidence beyond the horizon cannot rewrite history.

3.5 Manifest-Scoped Affine Calibration

The global calibrator uniquely minimizes natural-prevalence calibration NLL plus $10^{-6}(a^2 + b_{cal}^2)$ subject to $a \geq 10^{-6}$. Thus

$$P_e = \begin{cases} \sigma(az_e + b_{cal}) & e = 1, \kappa = \kappa_G, \\ \sigma(agz_e + b_g) & e = 1, \kappa = \kappa_g, \\ \text{null} & e = 0, \kappa = \kappa_\emptyset, \end{cases} \quad (8)$$

following positive-slope score calibration [4, 11]. The primary manifest supports only exact EDR+NDR mask 11, non-down/non-unknown health, valid parsers, resolved detector states available or no_detection, and exact pipeline versions. Any mismatch is out-of-support.

Before calibration, the manifest registers each group key as the RFC 8785/SHA-256 digest of a canonical object containing deployment profile and EDR/NDR detector versions; scenario, run, and label fields are forbidden. Conditional groups use deterministic balanced five-fold run-level cross-fitting on the calibration partition. Within each group, intact runs with binary counts (p_r, n_r) sort by decreasing $p_r + n_r$, then seeded SHA-256 and run ID. Each run goes to the fold minimizing the resulting maximum normalized positive, negative, and run loads; ties use total normalized load then fold ID. Every fold must be nonempty, and the ordered assignment map is sealed by digest. A group requires at least 50 positive revisions, 50 negatives, and 10 runs. For each held-out fold, improvement is $\Delta B = \text{Brier}_{global} - \text{Brier}_{group}$ and $\Delta N = \text{NLL}_{global} - \text{NLL}_{group}$, first macro-averaged by run and then equally across folds. Both means must be at least 0.01, every fold must be nonnegative, and paired run-bootstrap lower 95% bounds for both must exceed zero. A passing group is refit once on all calibration-partition records with that key; its final coefficients and key are sealed in the manifest. Manifest validation rejects duplicate keys. Runtime resolution is exclusive: first reject any support mismatch; otherwise select the single enabled exact key, or fall back to global when none is enabled. Scope Resolver alone emits g, κ , and $e = 1[n > 0 \wedge \kappa \neq \kappa_\emptyset]$; Calibrator only maps (z, e, κ, g) to P . This invariant prevents no_detection health support from making an empty detector vector score eligible.

3.6 Structured Sufficiency and Audit Assembly

$$U_e = (u_{coverage}, u_{provenance}, u_{lateness}, u_{contradiction}, u_{calibration}). \quad (9)$$

Table 2 makes every U axis a first-match deterministic rule rather than an informal confidence judgment.

The frozen contradiction rules are CR1, overlapping authenticated endpoint bindings map one endpoint key to different hosts; CR2, matched EDR socket and NDR flow disagree on the initiator after canonical direction normalization; and CR3, the same $(host, boot, pid, start)$ process key has conflicting parent/root lineage. A rule is applicable only when its relation exists; absent required fields then yield unevaluable. No applicable relation yields none. U is never scalarized and cannot gate, modify, or relabel P .

AuditRecord Assembler joins one EpisodeRevision, FeatureVector, z , SupportDecision, P , U , and the exact SourceCoverage objects referenced by SupportDecision, by $(episode_id, revision)$ and exact versions. It copies the health snapshot from those references; it cannot independently select or infer health. Required provenance comprises every membership event/root, selected link, scored input, coverage object, and version used by z, P, U ; only unscored diagnostic references are optional. Identical retries are idempotent; conflicting duplicates fail. At deterministic deadline $revision.watermark + 120$ s it atomically inserts exactly one AssemblyOutcome under that key: a complete AuditRecord, or an AssemblyFailureRecord with one code from $\{\text{assembly_timeout}, \text{missing_input}, \text{conflicting_duplicate}, \text{key_mismatch}, \text{version_mismatch}, \text{schema_invalid}, \text{digest_mismatch}\}$ and no fabricated score/probability fields. The unique-key store grants insert/select only and forbids a later success after terminal failure. The evaluation reports failure counts separately. Inputs are processed in fixed type order EpisodeRevision, FeatureVector, raw logit, SupportDecision, P , U , SourceCoverage. When multiple failures apply, precedence is schema, key, duplicate, digest, version, missing-input, then timeout. Before canonicalization, references sort by relation rank (membership/link/feature/health/contradiction), dependency-group UTF-8 bytes, object reference before sentinel, object-ID-or-empty bytes, then digest-or-empty bytes. RFC 8785 preserves JSON array element order rather than sorting it; JCS is applied only after this

Table 2: Deterministic U rules; rows list highest-to-lowest precedence.

Axis	Precedence	First matching predicate
coverage	unknown > unavailable > degraded > complete	missing/conflicting required interval; else any down/not-collected/parser-invalid; else one missed heartbeat or lag (120, 600] s; else all required intervals healthy
provenance	missing > partial > complete	unresolved scored input, membership event, selected link, coverage interval, or version; else optional unscored diagnostic absent; else all required roots resolve
lateness	late_accepted > revisable > stable	successor accepts late evidence; else open/finalized before fixed deadline; else closed
contradiction	present > unevaluable > none	any evaluable CR1–CR3 true; else an applicable rule lacks a required field; else all applicable rules false
calibration	out_of_support; supported_group; supported_global	exact serialization of Scope Resolver κ

application-level ordering. Optional absence uses a closed sentinel; missing required provenance fails assembly. Every closed VersionSet component binds a version identifier and SHA-256 digest for schema/parser/formation, detector registry, feature, normalizer, raw model, calibrator, support, sufficiency, assembly, and manifest. JSON numbers must be finite binary64 values represented by JSON decimal grammar; NaN, infinities, negative zero, and overflow fail schema validation. Runtime host, PID, wall-clock, and duration form a non-canonical envelope excluded from digests.

3.7 Boundary and Reproducibility Contract

The method excludes CTI, RAG/agents, impact/priority, automated response, and OT/ICS generalization. Appendix A.1 and the tables cited in the main text are normative. The contribution is limited to deterministic EDR/NDR episode construction, manifest-scoped probability, structured sufficiency, and replayable append-only provenance.

4 EVALUATION

The primary evaluation is an end-to-end BAS validation on isolated live Linux systems, not an offline event-generator experiment. CALDERA executes the attack families in Table 4; TRUSTEPISODE observes only EDR/NDR telemetry and authenticated source health. We evaluate whether the resulting episodes preserve attack evidence and whether the five evidence-sufficiency axes in Table 2 respond faithfully under adversarial telemetry perturbations and physical collector outages. Result cells and plots remain pending until the registered executions complete; no synthetic values or confidence intervals are inserted.

4.1 Research Questions and Evaluation Populations

RQ1 – live-system validity. Across the CALDERA scenarios in Table 4, does the label-blind builder recover the required attack actions without excessive fragmentation, over-merge, or benign contamination? **RQ2 – evidence sufficiency.** Do coverage, provenance, lateness, contradiction, and calibration states reflect the evidence actually available in each run without numerically altering the maliciousness probability? **RQ3 – adversarial robustness.** Under source masking, event loss, interval removal, delay, reordering, clock shift, duplication, and physical outage, how rapidly does performance degrade, which faults cause explicit sufficiency transitions, and which remain silent because authenticated health does not change? **RQ4 – replay and cost.** Are canonical outputs byte identical under replay, and what

throughput, latency, memory, storage, and recovery costs are observed?

The populations are disjoint views of the same registered runs. C_{form} contains one terminal revision per surviving episode lineage at $T_{form} = t_{run_end} + L_{wm} + W_{rev} = t_{run_end} + 1020$ s; merged parents are removed. C_{dec}^{all} contains the first finalized AssemblyOutcome per terminal lineage, including supported and out-of-support AuditRecords and AssemblyFailureRecords. The scorable inventory is

$$C_{dec}^{score} = \{o \in C_{dec}^{all} : o \text{ is AuditRecord, } o.decision_eligible, P \in [0, 1]\}.$$

After terminal insertion, the offline harness joins $L(o) \in \{positive, negative, mixed, ambiguous\}$ by EpisodeRevision digest. Class-dependent metrics use

$$C_{dec}^{binary} = \{o \in C_{dec}^{score} : L(o) \in \{positive, negative\}\}.$$

C_{stream} retains every revision-keyed terminal outcome for sufficiency transitions, replay equality, and cost. This separation prevents late revisions from duplicating RQ1/RQ2 candidates and prevents assembly failure from becoming hidden survivorship bias.

4.2 CALDERA-Based Live-System Validation

The primary testbed uses CALDERA v5.3.0 [1] as the BAS control plane, two isolated Ubuntu 22.04 targets, and an inline Suri-cata gateway. Scenario commands execute against the live process, file-system, service, and network state of the targets rather than against synthesized logs. An instrumented endpoint wrapper records generic process start/exit observations to an append-only spool; an independently stoppable forwarder emits label-free EDR events and authenticated health. The inline sensor supplies NDR observations. This is a controlled real-system validation, but it is not a production deployment or an evaluation of a commercial EDR product.

The BAS orchestrator executes frozen no-profile Bash abilities and records action receipts in a separate ground-truth store. CALDERA identifiers, ability metadata, receipts, and scenario labels are excluded from the online pipeline. Every run records public software versions plus image, configuration, parser, schema, topology, source, and hardware digests.

Table 3 specifies how the primary scenarios are analyzed. The unchanged execution is the paired reference for each replay or outage condition. All displayed states are placeholders until the live-system campaign completes.

Before a native run is sealed, the harness requires exactly one covering health interval from both EDR instances and the NDR instance at every event-time point. Physical-outage runs omit this

Table 3: Planned CALDERA live-system and adversarial-robustness analysis. Each condition is paired with the unchanged execution of the same scenario run. All result fields are pending.

Analysis block	Paired condition	Primary performance readout	Table 2 readout	Status
Native BAS execution	attack-only and concurrent-benign modes	action coverage; Frag.; Merge; Contam.; episode P/R/F1	five-axis state distribution at final revision	pending
Ranking/calibration	supported positive/negative outcomes	AUPRC; recall@10%; Brier; NLL; eligibility rate	calibration support and provenance completeness	pending
Replay integrity stress	delay, reorder, clock shift, duplicate	paired metric change; revision count; stabilization time	lateness/provenance transitions and contradiction rate	pending
Partial telemetry	source mask, random drop, interval removal	retained action coverage; eligibility; out-of-support rate	coverage/calibration transitions and silent-loss rate	pending
Physical outage	EDR/NDR stop for 120/600 s	detection, ready, and stable-recovery times	coverage, lateness, and calibration trajectories	pending
Deterministic replay	identical input at 1/2/4/8 workers	byte equality; throughput; p50/p95/p99; peak RSS	exact equality of all five sufficiency axes	pending

gate by design so that the authenticated gap remains observable. Runs fail closed when a manifest field is absent, a scenario action fails, or cleanup does not restore the target image.

4.3 Ground Truth, Matching, and Labels

Allowlist schemas reject `operation_id`, `ability_id`, adversary/technique labels, action receipts/windows, scenario class, and malicious labels from every online object [6]. Each ground-truth group g registers a nonempty set of required action tokens A_g ; each run registers benign/background tokens B_r . For predicted episode p , $A_{cov}(p, g) = |A_p \cap A_g|/|A_g|$, where A_p is its root-deduplicated attributed action set. Offline matching constructs the threshold-qualified edge graph E_q . An edge requires temporal IoU ≥ 0.10 , entity Jaccard ≥ 0.50 , and one exact offline receipt or registered action. Its weight is

$$w = 0.5A_{cov} + 0.3J_{entity} + 0.2IoU_{time}; \quad w \geq \theta_{match} = 0.50. \quad (10)$$

Maximum-weight one-to-one assignment over E_q operates within one run; ties use entity overlap, absolute time difference, then ascending bytes of $(episode_id, group_id)$. Assignment is used only for episode precision/recall/F1. Coverage, fragmentation, and overmerge use the unassigned graph.

A revision chooses the first incident edge under descending (A_{cov}, w) then ascending group ID. With no edge, $g^* = \text{null}$, $A_{cov} = 0$, and B_{cont} is undefined. Otherwise

$$B_{cont} = \frac{|A_p \cap B_r|}{|(A_p \cap A_{g^*}) \cup (A_p \cap B_r)|}.$$

First-match labels are mixed if $A_{cov} \geq \theta_{mal} = 0.50$ and $B_{cont} \geq \theta_{contam} = 0.25$; positive if coverage passes and contamination is below the threshold; negative if $A_{cov} = 0$, a registered benign token is present, and no conflict exists; otherwise ambiguous. Labels join only after the terminal outcome is inserted. Sensitivity varies $\theta_{mal} \in \{.25, .50, .75\}$, $\theta_{contam} \in \{.10, .25, .50\}$, and $\theta_{match} \in \{.40, .50, .60\}$ without changing the primary analysis.

4.4 Primary BAS Scenarios

Table 4 is the primary validation matrix. Each family is executed on the live Ubuntu targets in attack-only mode and with one deterministic concurrent benign workload. The protocol requires ten valid repetitions per family and mode; achieved counts and all measured outcomes remain pending.

B1–B8 cover Linux inventory, manifest-pinned package installation, file synchronization, registered service scanning, read-only

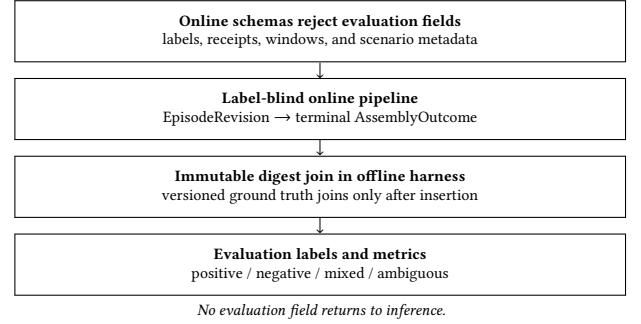


Figure 3: Ground Truth Firewall. Evaluation-only fields are rejected before inference and join immutable outputs only inside the offline harness.

SSH administration, log collection, configuration management, and supervisor-backed maintenance. Each benign card is run alone and as the registered concurrent background for malicious cards. Every card fixes preconditions, exact command templates, monotonic start offsets, fixed target paths, offline-only receipts, postconditions, cleanup, and failure predicates. Collection or action failure remains in the failure log but is excluded from performance estimates; cleanup failure blocks image reuse.

4.5 Adversarial Perturbations and Physical Outage

Each valid Table 4 execution has an immutable unchanged replay and deterministic adversarial children. Conditions are source masking; EDR- or NDR-specific random drop at 5/10/25%; removal of a 30/120/300 s interval; ingest delay of 30/120/600 s; reordering inside 30/120/600 s buffers; event-time shifts of $-120/-30/+30/+120$ s; and duplicate root groups at 5/10/25%. Other-source events and the original SourceCoverage journal remain fixed, so replay isolates data-integrity stress from health-reporting stress.

The perturbation seed is domain-separated by base run, operator, source, and dose. Before the operator is applied, base events sort by

$$(t^{event}, t^{ingest}, source_instance, event_id).$$

Duplicate IDs are derived from the sealed seed and original event ID; collisions invalidate the child. These rules make every adversarial condition exactly replayable without selecting an outcome after observing labels.

Table 4: Primary CALDERA BAS validation scenarios on live Ubuntu targets. Each card requires ten valid repetitions in attack-only and concurrent-benign modes; achieved counts are pending.

ID	Ordered CALDERA ability family	Stressor	Expected EDR/NDR evidence	Primary sufficiency axes
M1	discovery → peer inventory → SSH write	lateral chain	EDR lineage + NDR connection	provenance, contradiction
M2	stage → archive → SCP → delayed local action	weak-to-strong	EDR file/process + NDR transfer	lateness, provenance
M3	discovery → temporary-key SSH write	remote execution	EDR process + NDR session	coverage, calibration
M4	five-request 60 s cadence → local action	sparse sequence	NDR cadence + EDR action	coverage, lateness
M5	discovery → local configuration action	source-asymmetric case	EDR evidence; valid NDR no-detection	coverage, calibration
M6	matched-tool SSH → remote write	benign-tool overlap	contextual EDR/NDR evidence	contradiction, provenance
M7	actions at 0/300/3900 s	gap/span boundaries	boundary revisions and late evidence	lateness, provenance

Physical outage is evaluated separately because it changes authenticated source health rather than only event delivery. The cohort crosses source $\in \{\text{EDR, NDR}\}$, duration $\in \{120, 600\}$ s, and ten repetitions, balanced over M1/M3/M4/M7. After two healthy heartbeats, the selected collector is stopped and at least one scenario ability executes during the outage. We record health detection, first parser-valid ready interval after restoration, and the first time followed by 900 s without another revision. Detection and recovery are right-censored when their registered observation windows expire.

For run r , condition q , and scalar metric m , the paired effect is

$$\Delta m_{r,q} = m(r, q) - m(r, 0), \quad (11)$$

where $q = 0$ is the unchanged replay. For sufficiency axis k with categorical states a, b , the transition matrix is

$$T_{k,q}(a, b) = \frac{1}{N_q} \sum_{r=1}^{N_q} \mathbf{1}[U_k(r, 0) = a \wedge U_k(r, q) = b]. \quad (12)$$

For the action-coverage metric, the silent-loss rate is

$$SL_q = \frac{1}{N_q} \sum_{r=1}^{N_q} \mathbf{1}[A_{cov}(r, q) < A_{cov}(r, 0) \wedge U(r, q) = U(r, 0)]. \quad (13)$$

Robustness is therefore not defined as score invariance alone. Event loss with an unchanged authenticated health journal may leave $u_{coverage}$ unchanged by construction; the analysis reports such cases through Equation 13 rather than assigning an unsupported sufficiency transition. A desirable response can otherwise be stable episode evidence at low dose, graceful metric degradation at higher dose, or explicit movement to degraded, unavailable, or out_of_support when the probability is no longer justified. Silent confidence under missing evidence is treated as a failure.

4.6 Metrics and Statistical Reporting

For run r , let $E_{q,r}$ be its unassigned threshold-qualified graph. For each truth group g , $n_g = |\{p : (p, g) \in E_{q,r}\}|$; for episode p , $d_p = |\{g : (p, g) \in E_{q,r}\}|$. Formation reporting uses run-macro action coverage, $|G_r|^{-1} \sum_g \max(0, n_g - 1)$ fragmentation, $|P_r|^{-1} \sum_p \mathbf{1}[d_p > 1]$ over-merge, mean defined contamination, and one-to-one episode precision/recall/F1. Empty denominators are undefined, never zero.

Ranking and calibration use C_{dec}^{binary} only. AUPRC and recall among the top $\max(1, \lceil 0.10 |C_{dec,r}^{binary}| \rceil)$ per run are primary ranking metrics; Brier, NLL, and ten-bin reliability are reported only for supported records. AUROC and precision@ k are secondary [13]. For each Table 2 axis we report state counts and proportions, the

transition matrix in Equation 12, decision-eligible coverage, revision count, and time to stable closure. Equation 13 reports coverage loss not accompanied by any sufficiency transition. Outage plots report detection, ready, and stable-recovery time with censoring.

All summaries are macro-averaged over complete scenario runs. Paired perturbation effects use Equation 11; intervals use 2,000 percentile bootstrap resamples of complete run pairs with fixed seed 20260710 [3]. The paper will report the number of valid, failed, undefined, and censored runs beside every estimate. Deterministic-replay cost is measured at 1/2/4/8 workers using one warm-up and five measured repeats, with byte equality, throughput, ingest-to-outcome p50/p95/p99 latency, peak RSS, canonical storage bytes, and replay time.

4.7 External Entity-Level Context

Table 5 is secondary context, not the primary validation. It combines entity counts reported by CAPTAIN for KAIROS, NODLINK, and CAPTAIN on DARPA TC E3 CADETS/THEIA [2, 7, 14] with a deterministic TRUSTEPISODE projection. The external rows were not reproduced in this work, and the TRUSTEPISODE rows are an entity projection from episode membership rather than a native entity classifier. HOLMES is omitted because its reported grain is graph/campaign-level rather than an aligned E3 entity-count row [7, 8]. No cross-grain superiority claim follows.

For the reserved projection, an entity is predicted iff its canonical identifier belongs to a decision-eligible positive terminal episode; duplicates count once per evaluation cohort. A process-only scope aligns NODLINK, while an all-evaluable scope admits only entities in the frozen KAIROS/CAPTAIN-aligned mapping. Projection changes no formation, score, calibration, or threshold.

4.8 Pending Evidence and Limitations

The paper-facing CALDERA table and robustness figure intentionally retain pending markers until all scenario repetitions, perturbation pairs, and outage traces pass validation. Replacement values must include run counts, undefined/censored counts, and confidence intervals; a point estimate alone is insufficient. The controlled environment demonstrates execution against live operating systems and real EDR/NDR collection paths, but cannot establish production generalization, unseen-evasion robustness, or field base rates. Calibration claims apply only to exact supported manifests. The study excludes CTI, agentic response, impact/priority, OT/ICS, and autonomous remediation.

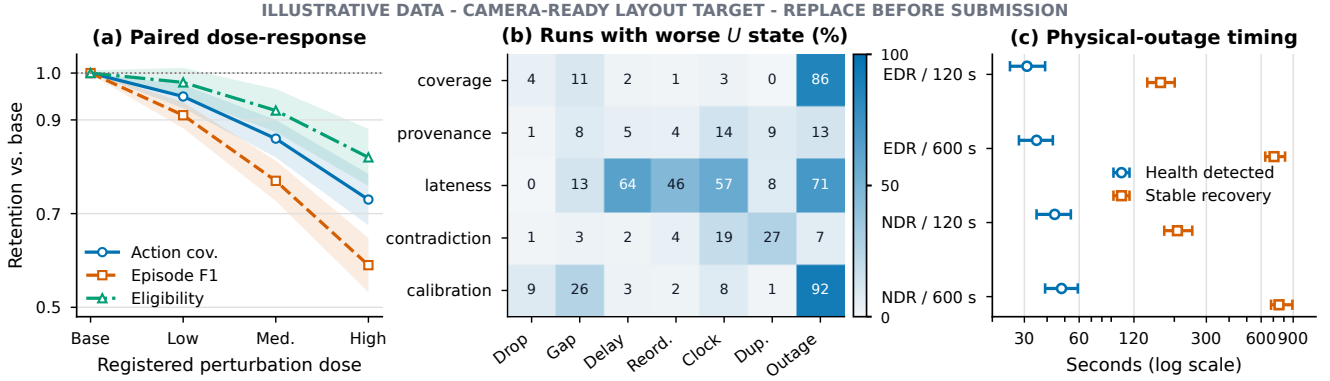


Figure 4: Illustrative camera-ready layout using simulated values, not experimental results. Panel (a) uses paired dose-response lines with 95% intervals; panel (b) uses a heatmap for the fraction of runs moving to a worse Table 2 state; panel (c) uses log-scale dot-and-whisker summaries for outage detection and stable recovery. All values must be replaced by registered measurements before submission; censored time-to-event curves belong in the supplementary material when censoring is material.

Table 5: Entity-level DARPA TC E3 context. KAIROS, NODLINK, and CAPTAIN counts are transcribed from CAPTAIN Table V [14] and were not reproduced in this work. TRUSTEPISODE is measured in this work using a deterministic projection from immutable episode membership. Precision, recall, and F1 are derived from the displayed counts. Bold denotes the highest precision, recall, or F1 within each dataset; ties are retained.

Dataset	Method	TP	FP ₀	FN	Precision	Recall	F1	Source
E3-CADETS	KAIROS	15	1,017	11	0.015	0.577	0.028	CAPTAIN Tbl V
	NODLINK	3	120	2	0.024	0.600	0.047	CAPTAIN Tbl V
	CAPTAIN	16	34	10	0.320	0.615	0.421	CAPTAIN Tbl V
	TRUSTEPISODE	16	64	10	0.200	0.615	0.302	this work
E3-THEIA	KAIROS	12	3,566	3	0.003	0.800	0.007	CAPTAIN Tbl V
	NODLINK	14	62	0	0.184	1.000	0.311	CAPTAIN Tbl V
	CAPTAIN	11	194	4	0.054	0.733	0.100	CAPTAIN Tbl V
	TRUSTEPISODE	10	9	5	0.526	0.667	0.588	this work

5 CONCLUSION

TRUSTEPISODE defines a narrow, replayable contract for EDR/NDR episode formation and candidate-conditioned maliciousness calibration. Episode Synthesis alone owns immutable membership revisions; the scorer, support resolver, calibrator, sufficiency evaluator, and AuditRecord Assembler have disjoint outputs; each revision terminates in either a complete AuditRecord or an explicit AssemblyFailureRecord; and missing or incompatible telemetry yields an explicit out-of-support state rather than a fabricated probability. The primary evaluation executes frozen CALDERA attack families on isolated live Ubuntu targets and pairs each execution with replay perturbations or authenticated collector outages. Performance deltas are interpreted together with the five sufficiency-state transitions, so honest abstention under missing evidence is distinguished from silent confidence. These CALDERA and robustness results remain pending; the current draft therefore makes no claim of deployment generalization.

A FROZEN SUBMISSION CONTRACTS

A.1 Machine Schema and Audit Assembly

The normative schema-ready in-PDF contract follows JSON Schema Draft 2020-12 closed-object semantics [15] for NormalizedEvent, SourceCoverage, EpisodeRevision, LateEvidenceRecord, FeatureVector, SupportDecision, AuditRecord, and AssemblyFailureRecord. FeatureVector owns D, C, m, o^{det} without health input. SupportDecision owns resolved a^{det}, κ, g , detector count n , eligibility reason, decision_eligible, EDR/NDR coverage_ref_ids, and health_snapshot_digest. Count $n = 0$ forces out-of-support, false eligibility, and null P even if both detectors report no_detection. AuditRecord requires those outputs, z, P, U , the revision digest, health snapshot, ordered references, exact versions, and canonical digest; out-of-support requires JSON null P and false eligibility. For each revision key the Assembler atomically inserts one terminal outcome: AuditRecord after a complete exact-version join, or AssemblyFailureRecord after conflict or deadline. The assembly deadline is revision.watermark + 120 s. The unique-key store is an insert/select sink and cannot later replace failure with success.

Identifiers are ASCII and sort by unsigned UTF-8 bytes. AuditRecord uses string IDs/digests, nonnegative integer revision, finite $D, P \in [0, 1]$, $C \in \{0, 1\}$, numeric z , Boolean eligibility, closed enums/maps for κ, U, m, h, a^{det} , ordered reference arrays,

and a closed version object. Only P and group key are nullable: out-of-support requires $P = \text{JSON null}$; only supported-group requires a non-null key. NaN, infinities, negative zero, and binary64 overflow are invalid. AssemblyFailureRecord contains no D, C, z, P, U and has exactly one of the seven failure codes in Section 3.6. Canonical bytes follow RFC 8785 JSON canonicalization [12]. JCS sorts object properties but preserves array order, so the application sorts every reference array first. SHA-256 includes each outcome’s declared digest-field set except canonical_digest. AuditRecord includes type/schema/key/revision, revision digest, $D, C, z, P, \kappa, g, e, U, m, h, o^{det}, a^{det}$, detector count/reason, ordered references, and versions; AssemblyFailureRecord includes type/schema/key/revision, revision digest, failure code, observed input types, and assembly-policy version; LateEvidenceRecord includes type/schema/episode/latest revision and digest/event, arrival/deadline/reason, raw pointer/root, ordered lineage references, and VersionSet. Host, PID, wall time, and duration are an optional non-canonical envelope. References sort by relation rank, dependency-group bytes, object-before-sentinel, object-ID-or-empty bytes, then digest-or-empty bytes. Optional absence uses a closed sentinel; required absence fails assembly. Assembly input order is revision, feature, logit, support, probability, U , then coverage. Failure precedence is schema, key, duplicate, digest, version, missing, then timeout. Every VersionSet component closes both a version identifier and content digest for schema, parser, formation, detector registry, feature, normalizer, model, calibrator, support, sufficiency, assembly, and manifest. Conforming validation must accept supported, unsupported, late, and assembly-failure records and reject any online record containing malicious_label. Online objects reject all label, scenario, marker/window, and ability/operation fields.

A.2 Episode Algorithm and State Machine

Fixed values are $W_{gap} = 300$ s, $W_{max} = 3600$ s, $L_{wm} = 120$ s, $W_{rev} = 900$ s, and $W_{ret} = 86400$ s. Relations have $\rho = (4, 3, 2, 1)$ for causal/session/endpoint/entity and n_R counts satisfied classes; candidates sort by $(-\rho, -n_R, \Delta t, t_{start}, id)$. A hub has at least 25 relations in 600 s. Merge permits at most three parents connected by causal, session, or endpoint relations and at most 500 events/200 entities; otherwise attach to the first candidate. The event order is

$$(t^{event}, t^{ingest}, source_instance, event_id).$$

First finalization freezes t_f and deadline $t_f + 900$ s. The lineage then leaves the open index and remains in a revisable index; at closure, canonical relation keys remain in a closed-lineage index through deadline plus W_{ret} . Each event queries revisable/closed indexes first and returns after exactly one successor/LateEvidenceRecord; only no match permits open start/attach/merge. A qualifying late event before the fixed deadline creates a finalized successor without reopening or extending the horizon; the deadline creates a membership-identical closed successor. A later related event creates only a LateEvidenceRecord in the late store with reason beyond_revision_horizon, closed_lineage_match, or duplicate_after_terminal_outcome. Replay equality covers every state, late record, output, reference, and digest.

A.3 Thresholds and Sufficiency Rules

Table 6 is the single numeric and precedence registry for the submission PDF.

Feature Extractor emits only detected, examined_none, or not_examined. At EpisodeRevision.event_time_end, Scope Resolver combines that observation with exactly one content-distinct covering SourceCoverage interval per expected instance and emits available, no_detection, collector_unavailable, parser_invalid, or unknown; the first two are health-support states, but zero eligible detector values still force out-of-support. CR1 is an overlapping endpoint-to-host binding conflict, CR2 is canonical EDR/NDR flow-direction conflict, and CR3 is conflicting parent/root lineage for one process key. Health is healthy for zero missed 30 s heartbeats, lag ≤ 120 s, valid parser, and live transport; degraded for one missed heartbeat or lag $(120, 600]$ s; and down for two misses or authenticated stop. Required provenance is every membership/root, selected link, scored input, coverage object, and version; diagnostics are optional. Any later executable threshold file must exactly match Table 6.

A.4 Calibration Resolver and Invalidation

The global fit uniquely minimizes natural-prevalence NLL plus $10^{-6}(a^2 + b^2)$ with $a \geq 10^{-6}$. A registered group key is the RFC 8785/SHA-256 digest of deployment profile and EDR/NDR detector versions; labels, scenario, and run are forbidden. Five-fold calibration-partition cross-fitting keeps each run intact. For each group, runs with binary counts (p_r, n_r) sort by decreasing $p_r + n_r$, then SHA-256 of seed/group/run. Each run is greedily assigned to the fold minimizing the resulting maximum normalized positive, negative, and run loads, then total normalized load and fold ID; the assignment map is sealed and hashed. The gate uses positive improvements $\Delta B = B_{global} - B_{group}$ and $\Delta N = NLL_{global} - NLL_{group}$. It evaluates the gate, after which a passing group is refit on all calibration records for that key and sealed with its coefficients. Manifest validation rejects duplicate keys. Runtime resolution first checks exact mask 11, health, parser, resolved detector states, and formation/feature/model/label/schema/pipeline versions; mismatch yields out-of-support. Otherwise one enabled exact group yields supported_group, and no enabled key falls back to supported_global. Scope Resolver alone emits the key, status, and eligibility; Calibrator only emits P . Episode/candidate, feature/normalizer, raw-model, or label changes invalidate calibration; additive unused fields require equivalence proof.

A.5 Live-System Lab, Cards, and Offline Labels

The lab, card, and label contract is exactly Sections 4.2–4.4, Equation 10, and Table 4. It pins public CALDERA and OCSF versions; missing manifest fields fail closed; each card/mode requires ten valid runs with achieved counts pending. Each Linux-only M/B card closes image key and digest, ordered ability IDs, no-profile Bash executor plus exact command templates, monotonic offsets, fixed target paths, offline receipt preimage, success/failure predicates, cleanup commands and digest check, and allowed mode. Target commands cannot receive run/card/ability/operation/receipt identifiers or values; orchestrator-only receipts remain in Ground

Table 6: Frozen numeric registry and deterministic precedence.

Contract owner	Parameter(s)	Frozen value / order	Validation rule
A. Numeric contract			
Formation	W_{gap} ; W_{max} ; L_{wm} ; W_{rev} ; W_{ret}	300; 3600; 120; 900; 86400 s	sealed before final training fit
Formation guards	hub relations/window; parents; component cap	25/600 s; 3; 500 events/200 entities	reject on violation; never learned
Feature	detector IDs; top- k ; source mask	3 EDR + 3 NDR; $k = 3$; EDR+NDR	finite $[0, 1]$; mid-rank CDF
Offline matching	θ_{mal} ; θ_{contam} ; θ_{match}	.50; .25; .50	freeze before label release
Match predicates	temporal IoU; entity Jaccard	.10; .50	also require exact action/receipt
Group gate	positives; negatives; runs; folds	50; 50; 10; 5	deterministic balanced run folds
Gate gain	mean Δ Brier; mean Δ NLL; uncertainty	$\geq .01$; $\geq .01$; lower 95% > 0	every held-out fold nonnegative
Uncertainty	bootstrap repetitions; seed	2000; 20260710	resample complete run clusters
B. Deterministic U precedence (first match wins)			
$u_{coverage}$	health/parser state	unknown $>$ unavailable $>$ degraded $>$ complete	evaluate authenticated coverage first
$u_{provenance}$	required references	missing $>$ partial $>$ complete	required roots precede optional diagnostics
$u_{lateness}$	revision state	late_accepted $>$ revisable $>$ stable	accepted successor; before deadline; closed
$u_{contradiction}$	CR1–CR3	present $>$ unevaluable $>$ none	first applicable conflict rule wins
$u_{calibration}$	exact support enum	out_of_support; supported_group; supported_global	exact manifest and support resolution

Truth Store. Template variables resolve only from the frozen run manifest. Malicious modes are attack-only or one hash-selected B1–B8 workload begun 60 s earlier; benign cards also run standalone. Missing values and malformed inputs are counted, not zero-filled.

A.6 Adversarial Robustness and Statistical Contract

Formation coverage, fragmentation, and over-merge use the threshold-qualified graph; one-to-one assignment is restricted to episode precision/recall/F1. Decision counts use C_{dec}^{all} ; scorable coverage uses C_{dec}^{score} ; class-dependent metrics use positive/negative C_{dec}^{binary} . Outcome and failure labels join only after terminal insertion. Source mask, drop, interval removal, delay, reorder, clock shift, and duplication instantiate every registered EDR/NDR source and dose specified in Section 4.5. Every condition is paired with the immutable unchanged replay of the same live-system run.

The five axes in Table 2 are reported as categorical state counts, proportions, and paired transition matrices. Scalar changes follow Equation 11; transitions follow Equation 12. The physical-outage cohort contains exactly 40 traces (two sources by two durations by ten repetitions), balanced over M1/M3/M4/M7, and fixes health detection, ready, stable-recovery, follow-up, and censoring rules. Metrics, run-level resampling, undefined/censored reporting, and deterministic replay cost are exactly Sections 4.1, 4.5, and 4.6 plus Table 3. All measured fields remain pending until the registered executions complete.

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GENERATIVE AI USE DISCLOSURE

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